

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09

COURSE NAME: Programming in Java COURSE

OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) Total Marks: 100

ANNEXURE A

Assignment 1

Code of Conduct:

I B. Priyanka (192421206) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

Annexure A

Assignment 1

Question 24 – Online Auction System

Problem Statement:

An auction website records bidder information.

Questions

- a) Develop a Java program using **PriorityQueue**.
 - b) Display the highest bidder.
 - c) Compare Queue and PriorityQueue.
 - d) Explain priority scheduling.
-

CODE:

```
import java.util.PriorityQueue;

public class OnlineAuctionSystem {

    static class Bidder implements Comparable<Bidder> {

        String name;

        int bidAmount;

        Bidder(String name, int bidAmount) {

            this.name = name;

            this.bidAmount = bidAmount;

        }

        public int compareTo(Bidder other) {

            return other.bidAmount - this.bidAmount;

        }

    }

    public static void main(String[] args) {
```

```

PriorityQueue<Bidder> auctionQueue = new PriorityQueue<>();
auctionQueue.add(new Bidder("Arun", 5000));
auctionQueue.add(new Bidder("Priya", 8000));
auctionQueue.add(new Bidder("Kumar", 6500));
auctionQueue.add(new Bidder("Ravi", 7000));
auctionQueue.add(new Bidder("Anjali", 9000));
auctionQueue.add(new Bidder("Suresh", 7500));
auctionQueue.add(new Bidder("Meena", 10000));
auctionQueue.add(new Bidder("Rahul", 8500));
Bidder highestBidder = auctionQueue.peek();
System.out.println("Highest Bidder:");
System.out.println("Name: " + highestBidder.name);
System.out.println("Bid Amount: Rs." + highestBidder.bidAmount);
System.out.println("\nAll Bidders in Priority Order:");
while (!auctionQueue.isEmpty()) {
    Bidder bidder = auctionQueue.poll();
    System.out.println("Name: " + bidder.name +
        ", Bid Amount: Rs." + bidder.bidAmount);
}
}
}

```

QUESTIONS:

a) Develop a Java program using **PriorityQueue**

```

import java.util.PriorityQueue;

public class OnlineAuctionSystem {
    static class Bidder implements Comparable<Bidder> {
        String name;
        int bidAmount;
        Bidder(String name, int bidAmount) {
            this.name = name;

```

```

        this.bidAmount = bidAmount;
    }

    public int compareTo(Bidder other) {
        return other.bidAmount - this.bidAmount;
    }
}

public static void main(String[] args) {
    PriorityQueue<Bidder> auctionQueue = new PriorityQueue<>();
    auctionQueue.add(new Bidder("Arun", 5000));
    auctionQueue.add(new Bidder("Priya", 8000));
    auctionQueue.add(new Bidder("Kumar", 6500));
    auctionQueue.add(new Bidder("Ravi", 7000));
    auctionQueue.add(new Bidder("Meena", 10000));
    Bidder highestBidder = auctionQueue.peek();
    System.out.println("Highest Bidder: " + highestBidder.name);
    System.out.println("Highest Bid Amount: Rs." + highestBidder.bidAmount);
}
}

```

b) Highest Bidder

The highest bidder is **Meena** with a bid amount of **Rs.10000**.

c) Queue vs PriorityQueue

Queue	PriorityQueue
Follows FIFO order	Follows priority order
First element is removed first	Highest-priority element is removed first
Used for normal processing	Used for priority-based processing

d) Priority Scheduling

Priority scheduling is a method in which the element with the **highest priority is processed first**. In this auction system, the bidder with the **highest bid amount** is given the highest priority.

OUTPUT:

OnlineAuctionSystem.java

Share

Run

```
1- import java.util.PriorityQueue;
2- public class OnlineAuctionSystem {
3-     static class Bidder implements Comparable<Bidder> {
4-         String name;
5-         int bidAmount;
6-         Bidder(String name, int bidAmount) {
7-             this.name = name;
8-             this.bidAmount = bidAmount;
9-         }
10-        public int compareTo(Bidder other) {
11-            return other.bidAmount - this.bidAmount;
12-        }
13-    }
14-    public static void main(String[] args) {
15-
16-        PriorityQueue<Bidder> auctionQueue = new PriorityQueue<>();
17-
18-        auctionQueue.add(new Bidder("Arun", 5000));
19-        auctionQueue.add(new Bidder("Priya", 8000));
20-        auctionQueue.add(new Bidder("Kumar", 6500));
21-        auctionQueue.add(new Bidder("Ravi", 7000));
```

Output

Highest Bidder:
Name: Meena
Bid Amount: Rs.10000

All Bidders in Priority Order:
Name: Meena, Bid Amount: Rs.10000
Name: Anjali, Bid Amount: Rs.9000
Name: Rahul, Bid Amount: Rs.8500
Name: Priya, Bid Amount: Rs.8000
Name: Suresh, Bid Amount: Rs.7500
Name: Ravi, Bid Amount: Rs.7000
Name: Kumar, Bid Amount: Rs.6500
Name: Arun, Bid Amount: Rs.5000

=== Code Execution Successful ===